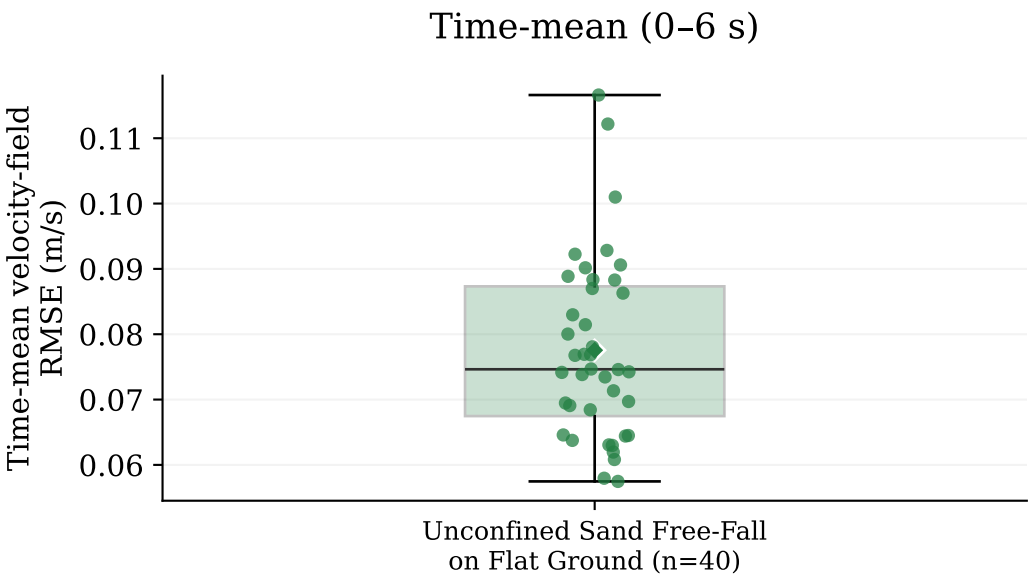
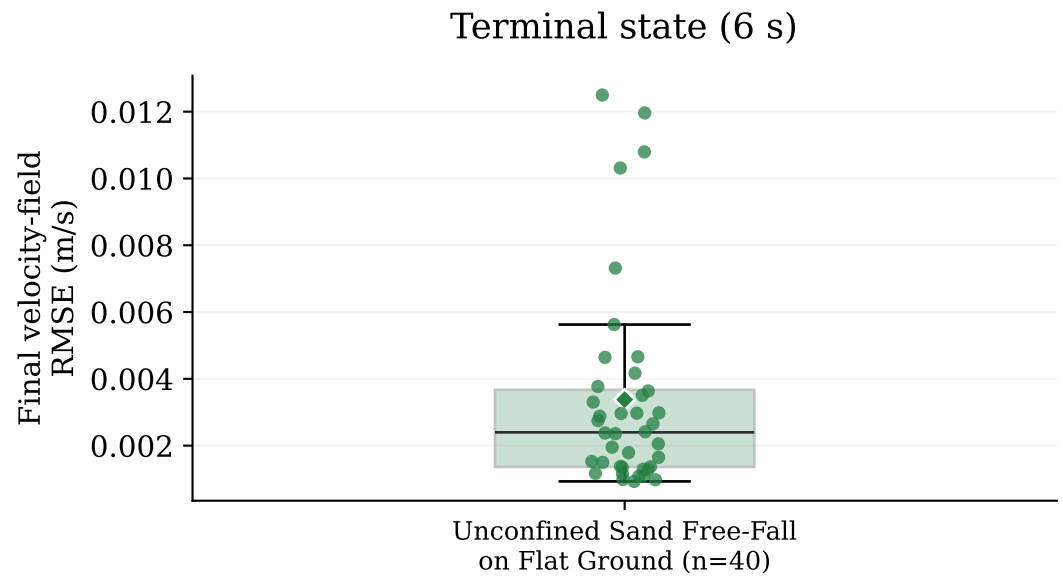
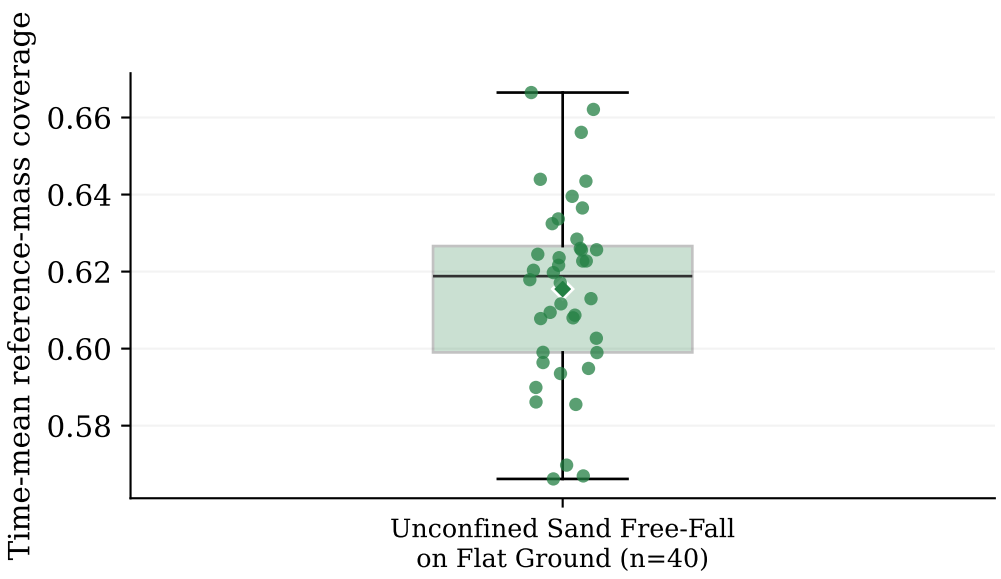
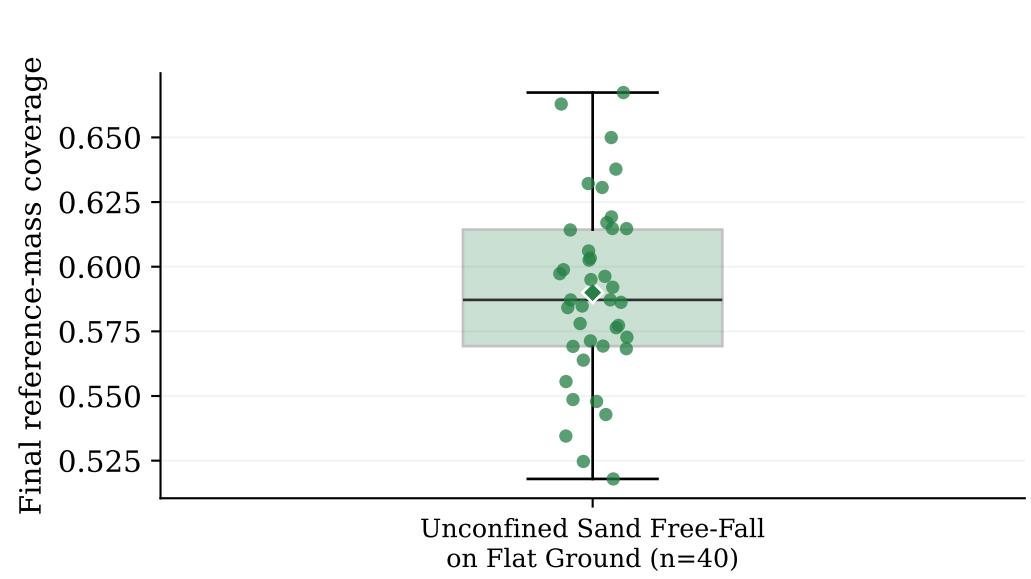
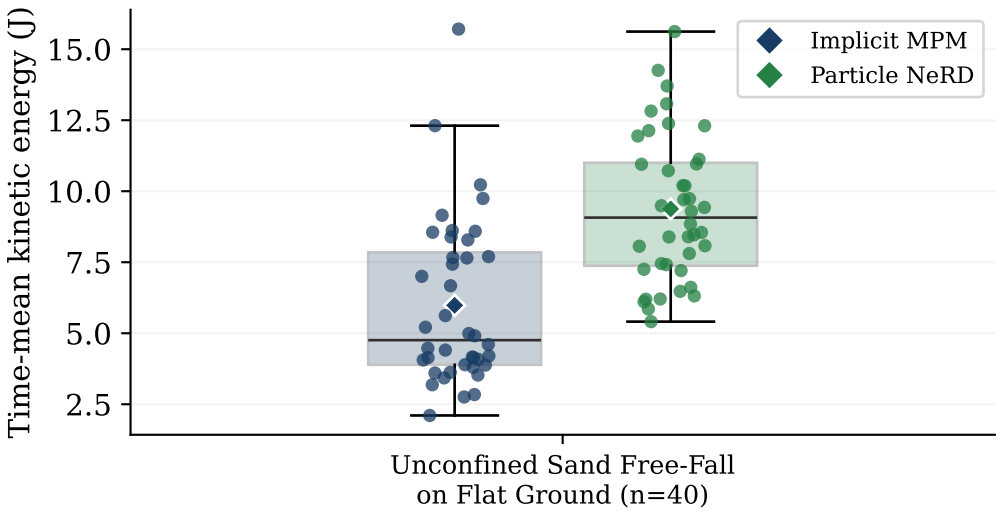
Group C: Unconfined Sand Free-Fall Motion on Flat Ground (No Rigid Object, n=40)











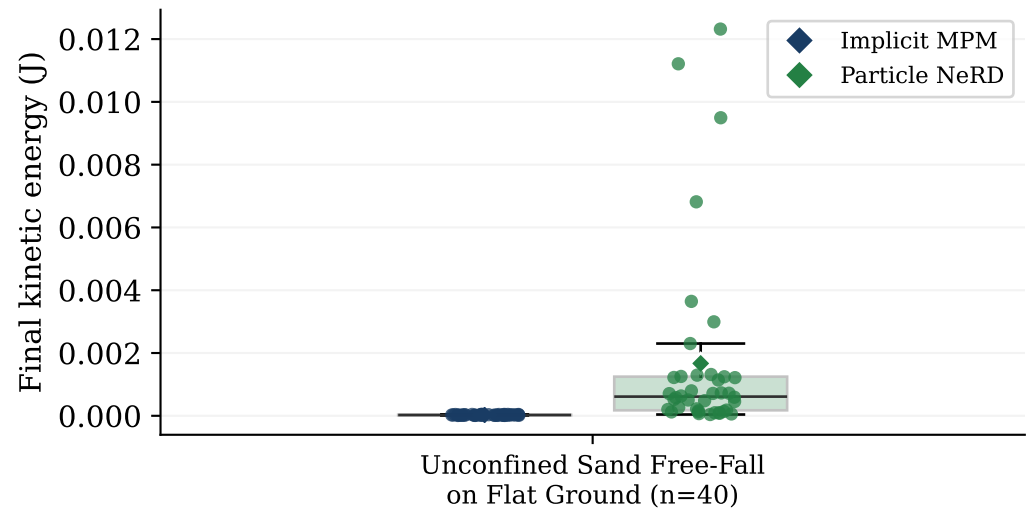


Figure 4c: Frozen object-model checkpoint 11,000. Velocity-field RMSE, reference-mass coverage, and particle kinetic energy for unconfined 3D sand blocks/columns dropped onto open flat ground (n=40).